Fiscal Year:	FY 2008	**Task Last Updated:**	FY 09/24/2008
PI Name:	Egolfopoulos, Fokion Ph.D.		
Project Title:	Quantitative Studies on the Propagation and Extinction of Near-Limit Premixed Flames Under Normal and Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	90089-1453	**Congressional District:**	33
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date extended to 6/30/2008 per PI (10/07) NOTE: Received NCE to 9/30/2007 per PI (10/06)		
Key Personnel Changes/Previous PI:	Yufei Dong ; Adamandios Sifounakis		
COI Name (Institution):	Spedding, Geoff (University of Southern California)		
Grant/Contract No.:	NNC04AA11A		
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Task Description:	The objectives of this program are twofold. The first is to utilize the results of the previous NASA support programs to continue to produce reference experimental data on near-limit flame propagation and extinction that will be of fundamental and practical importance. All experiments were performed for atmospheric flames in the counterflow configuration. The second objective is to compare the experimental data against numerical predictions and through such comparisons to validate current models describing the processes of chemical kinetics and thermal radiation and also gain insight into the mechanisms controlling the response of near-limit flames. Task Objective:		

	To obtain quantitative information about the characteristics of weakly burning flames in microgravity, and to use this information to analyze near-limit phenomena at normal gravity and microgravity conditions. The research will allow for the study of weakly burning flames without the complications of buoyancy, wall heat loss or time varying strain. It will provide fundamental data about the flammability limits, and has applications to turbulent combustion.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	It will provide fundamental data about the flammability limits, and has applications to turbulent combustion and space fire safety.
Task Progress:	Progress was made in four areas: 1. A new experimental technique was developed for the measurement of laminar flame speeds by establishing "cake" flames in the counterflow configuration. The cake flames are characterized by very small Karlovitz numbers and the obtained results are close to previous measurements. This approach may be appropriate for use in micro-gravity facilities. 2. The laminar flame speed has been traditionally used for the partial validation of flame kinetics. In most cases, however, its accurate determination requires extensive data processing and/or extrapolations, thus rendering the measurement of this fundamental flame property indirect. Additionally, the presence of flame front instabilities does not conform to the definition of the laminar flame speed. This is the case for Le < 1 flames, with the most notable example being the ultra-lean H2/air flames, which develop cellular structures at low strain rates so that the determination of laminar flame speeds for such mixtures is not possible. Thus, this low-temperature regime of H2 oxidation has not been validated systematically in flames. In the present investigation, an alternative/supplemental approach is proposed, which includes the experimental determination of extinction strain rates for these flames and comparing them with the predictions of direct numerical simulations. This approach is meaningful for two reasons: (1) extinction strain rates can be measured directly, as opposed to laminar flame speeds, and (2) while the unstretched lean H2/air flames are cellular, the stretched ones are not, making thus comparisons between experiment and simulations meaningful. Such comparisons revealed serious discrepancies between experiments and simulations for ultra-lean H2/air flames by using four kinetics mechanisms. Additional studies were conducted for lean and near-stoichiometric H2/air flames diluted with various amounts of N2. Similarly to the ultra-lean flames, significant discrepancies between experimental and predicted extinction strain rates were also found. In order to identify the possible sources of such discrepancies the effect of uncertainties on the diffusion coefficients was assessed and an improved treatment of diffusion coefficients was advanced and implemented. Under the conditions considered in this study, the sensitivity of diffusion coefficients on the extinction response was found to be significant and for certain species more so than the kinetics rate constants. 3. The lean flammability limits of CH4/air and C3H8/air mixtures were numerically determined for a wide range of pressures and unburned mixture temperatures in order to assess the near-limit flame behavior under conditions of relevance to internal combustion engines. The study included the simulation of freely propagating flames with the inclusion of detailed descriptions of chemical kinetics and molecular transport, radiative loss, and a one-point continuation method to solve around singular points as the flammability limit is approached. Results revealed that both pressure and unburned mixture temperature have significant effects on the lean flammability limit as well as the attendant limit flame temperature. Specifically, the lean limit was found to first increase and then decrease with pressure, while the limit temperature decreases with pressure in general, and can be reduced to values as low as 900 K under engine-like conditions. Through sensitivity and species consumption path analyses it was further shown that the chain mechanisms that control the near-limit flame response critically depend on the thermodynamic state of the mixture. Thus, mechanisms that are identified as important at near-atmospheric conditions may not be relevant at higher pressures and unburned mixture temperatures. In particular, the response of near-limit flames was found to resemble the homogeneous explosion limits of hydrogen/oxygen mixtures in that while at low pressures the main branching and termination reactions are respectively H + O2 --> OH + O and H + O2 +M --> HO2 +M, at the elevated pressures relevant to internal combustion engines the system branching is controlled by the HO2-H2O2 kinetics. Potential avenues for extending the lean operation limits of internal combustion engines are suggested based on the understanding gained herein. 4. The flame stability was assessed under normal gravity by using variable density inert gases. The hypothesis was that even under normal gravity flame stability could be enhanced for near-limit flames if the density gradients are minimized through the use of light inert gases such as helium whose density is seven times less than air and nearly equal to the density of typical combustion products. The studies were conducted in the opposed-jet configuration by counterflowing helium exiting the top burner against a hydrocarbon/air jet exiting the bottom burner. Helium was used also as the ambient gas. Experiment confirmed that by counterflowing methane/air or propane/air flames against helium extends the flame stability limits compared to using air instead of helium. This is due to the fact that by using helium the density gradients downstream of the flame are nearly eliminated and light gas always resides over heavy gas. The H2/CO/C1-C4 kinetic model (111 species and 784 reactions), applicable to a wide variety of combustion scenarios, incorporates the recent thermodynamic, kinetic, and species transport updates relevant to high-temperature oxidation of hydrogen, carbon monoxide, and C1-C4 hydrocarbons was developed. [Wang, H., You, X., Joshi, A.V., Davis, S.G., Laskin, A., Egolfopoulos, F.N. & Law, C.K. (2007) USC Mech Version II. High-Temperature Combustion Reaction Model of H2 /CO/C 1 -C 4 Compounds. http://ignis.usc.edu/]
Bibliography Type:	Description: (Last Updated: 10/19/2009)
Articles in Peer-reviewed Journals	Egolfopoulos FN, Holley AT, Law CK. "An assessment of the lean flammability limits of CH4/air and C3H8/air mixtures at engine-like conditions." Proceedings of the Combustion Institute, 2007 Jan;31(2):3015-22. http://dx.doi.org/10.1016/j.proci.2006.08.018 , Jan-2007